Do the following exercises from Chapter 5 of the textbook: 2, 3, 5, and 7. Can scan in information as appropriate and add to the MS Word document for submission.

1.

2. Calculate the direct and total direct costs of labour for a project team member using the following data:

Hourly rate: $50/hr

Hours needed: 100

Overhead rate: 30%

Ans: Direct Labor Cost:

$$Direct\ Labor\ Cost\ =\ Hourly\ Rate\ \times\ Hours\ Needed$$

Hourly Rate = $50/hr

Hours Needed = 100 hours

$$Direct\ Labor\ Cost\ =\ 50\ \times\ 100\ =\ 5000$$

So, the direct labour cost is $5000.

Total Direct Cost (Including Overhead):

$$Overhead\ Cost\ =\ Direct\ Labor\ Cost\ \times\ Overhead\ Rate$$

Overhead Rate = 30% = 0.30

$$Overhead\ Cost\ =\ 5000\ \times\ 0.30\ =\ 1500$$

So, the overhead cost is $1500.

$$Total\ Direct\ Cost\ =\ Direct\ Labor\ Cost\ +\ Overhead\ Cost\ =\ 5000\ +\ 1500\ =\ 6500$$

2.

3. The Munsters have been saving money in order to buy a house. They figure that, given current interest rates, they could afford a $400,000 home. Before looking at houses on the market they decide to explore the possibility of building a new home. The Munsters figure they could buy a suitable lot for $70,000–$75,000. At a minimum they want to build a 2,400-square-foot house. The cost for a house of the quality they desire is $160 per square foot. Given this information, should the Munsters pursue the option of building a new house?

Ans:

Cost of the Lot:

The cost of the lot is between $70,000 and $75,000. For calculation purposes, we will use the average of these two values:

$$Cost\ of\ Lot\ =\ \frac{70\,,000\ +\ 75\,,000}{2}\ = 72\,,500$$

Minimum Cost of lot is 70,000

Cost to Build the House:

The Munsters want to build a 2,400-square-foot house, and the construction cost is $160 per square foot. So, the cost to build the house is:

$$Cost\ to\ Build\ House\ =\ 2\,,400\ sq\ ft\ \times\ 160\ per\ sq\ ft\ =\ 384{,}000$$

Total Cost to Build the House:

To find the total cost of building the new house, we add the cost of the lot to the cost of building the house:

$$Total\ Cost\ to\ Build\ =\ Cost\ of\ Lot\ +\ Cost\ to\ Build\ House$$

$$Total\ Cost\ to\ Build\ =\ 72{,}500\ +\ 384{,}000\ =\ 456{,}500$$

$$Minimum\ Total\ cost\ to\ build\ a\ house\ =\ 70{,}000\ +\ 384{,}000\ =\ 454{,}000$$

The total cost to build a new home ($456,500) or a minimum of $454,000 which exceeds their budget of $400,000. **So, based on the current information, the Munsters should not pursue the option of building a new house,** as it would exceed their budget. They may need to either lower the square footage of the house or consider other options like looking for an existing home within their price range.

3.

5. Exercise Figure 5.1 is a project WBS with cost apportioned by percentages. If the total project cost is estimated to be $800,000, what are the estimated costs for the following deliverables?

a. Design

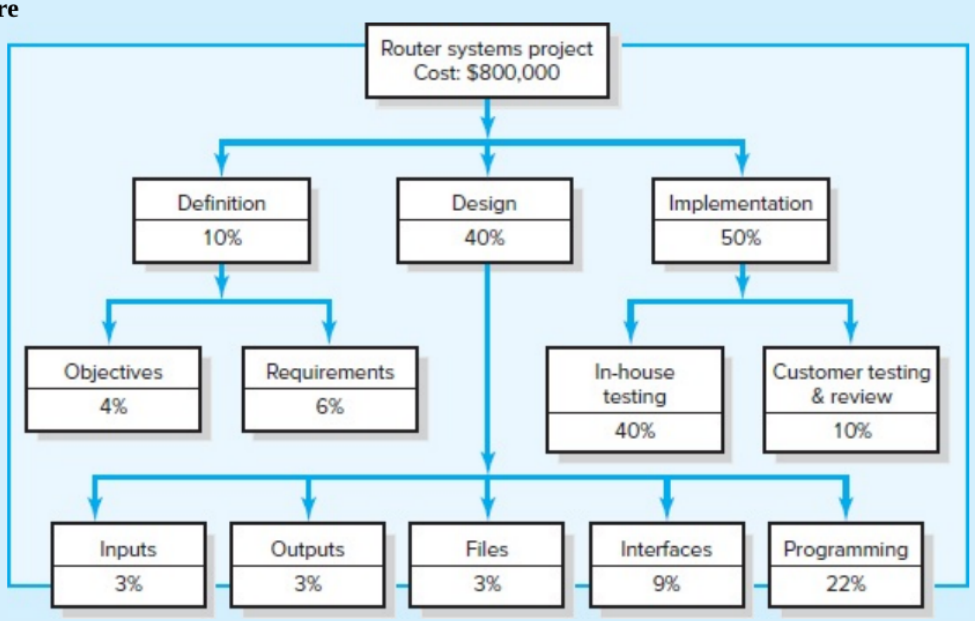
b. Programming

c. In-house testing

What weaknesses are inherent in this estimating approach?

EXERCISE FIGURE 5.1

WBS Figure



Ans:

a. Design

The WBS shows that Design is allocated 40% of the total project cost.

$$Design\ Cost = 800{,}000 \times 40\% = 800{,}000 \times 0.40 = 320{,}000$$

So, the estimated cost for Design is $320,000.

b. Programming

The WBS shows that Programming is allocated 22% of the total project cost.

$$Programming\ Cost = 800{,}000 \times 22\% = 800{,}000 \times 0.22 = 176{,}000$$

So, the estimated cost for Programming is $176,000.

c. In-house testing

The WBS shows that In-house testing is allocated 40% of the Implementation phase, and the Implementation phase is allocated 50% of the total project cost.

First, we need to calculate the total amount allocated for In-house testing:

$$In - house\ testing\ Cost = 800{,}000 \times 50\% \times 40\% = 800{,}000 \times 0.50 \times 0.40 = 160{,}000$$

Or

$$In - house\ testing\ = \$800{,}000 \times 0.40 = \$320{,}000$$

So, the estimated cost for In-house testing is $320,000.

Weaknesses in This Estimating Approach:

Over-simplification: The WBS assigns fixed percentages to broad project categories without considering the varying complexity of individual tasks. This can lead to inaccurate estimates, as some activities may require significantly more effort, expertise, or resources than others.

Lack of Detail: The percentage-based approach lacks sufficient granularity. It doesn't provide insight into the specific resources, time, or effort needed for each task, making it challenging to adjust estimates as the project evolves.

Rigid Assumptions: This method relies on static percentage allocations, assuming they remain valid throughout the project. However, as the project progresses and scope or requirements change, these assumptions may no longer hold true.

Ignoring Risks and Uncertainties: The approach doesn't factor in potential risks or the need for contingency planning. It fails to account for unexpected issues like delays or resource constraints, which can significantly impact project cost and timelines.

4.

7. Omega 2 Project. Using the “complexity weighting” scheme shown in Table

5.2 and the following function point complexity weight table, estimate the

total function point count. Assume historical data suggest five function points equal

one person a month and six people have been assigned to work on the project.

TABLE 5.2

Simplified Basic Function Point Count Process for a Prospective Project or Deliverable

	Complexity Weighting			
Element	**Low**	**Average**	**High**	**Total**
Number of *inputs*	____ × 2 +	____ × 3 +	____ × 4	= ____
Number of *outputs*	____ × 3 +	____ × 6 +	____ × 9	= ____
Number of *inquiries*	____ × 2 +	____ × 4 +	____ × 6	= ____
Number of *files*	____ × 5 +	____ × 8 +	____ × 12	= ____
Number of *interfaces*	____ × 5 +	____ × 10 +	____ × 15	= ____

Complexity Weight Table		
Number of inputs	15	Rated complexity low
Number of outputs	20	Rated complexity average
Number of inquiries	10	Rated complexity average
Number of files	30	Rated complexity average
Number of interfaces	50	Rated complexity high

a. What is the estimated project duration?

b. If 20 people are available for the project, what is the estimated project duration?

c. If the project must be completed in six months, how many people will be needed for the project?

Ans:

$$Inputs\ (15, low)\ \times\ 2\ =\ 15 \times 2 = 30$$

$$Outputs\ (20, average)\ \times\ 6\ =\ 20 \times 6 = 120$$

$$Inquiries\ (10, average)\ \times\ 4\ =\ 10 \times 4 = 40$$

$$Files\ (30, average)\ \times\ 8\ =\ 30 \times 8 = 240$$

$$Interfaces\ (50, high)\ \times\ 15\ =\ 50 \times 15 = 750$$

$$Total\ Function\ Points\ =\ 30 + 120 + 40 + 240 + 750 = 1180$$

Historical conversion:

$$Person - months\ =\ \frac{1180}{5}\ =\ 236$$

a. 6 people assigned

$$Duration\ with\ 6\ people\ =\ \frac{236}{6}\ \approx 39.33\ months$$

b. 20 people assigned

$$Duration\ with\ 20\ people\ =\ \frac{236}{20}\ \approx\ 11.8\ \ months$$

c. If the project must be completed in 6 months

$$Number\ of\ people\ needed\ = \frac{236}{6} \approx 39.33$$